

Lean Product Requirements Document: Jetty Planning & Monitoring System

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1. Product Overview

- **Vision / Goal:** To digitize and streamline end-to-end jetty operations by providing real-time visibility, automated SLA calculations, and standardized workflows for vessel loading/unloading.
- **Target Users:** Jetty Operators, Logistics & EXIM teams, Quality Control (QC), Tank Farm teams, and PPIC for monitoring.

2. Problem Statement

Jetty operations currently lack a centralized system for tracking vessel progress, leading to manual SLA calculations, fragmented communication between departments (QC, Tank Farm, Jetty), and potential demurrage costs due to inefficient planning.

3. Success Metrics

- **SLA Accuracy:** Variance between system-calculated Estimated Completion and actual completion time.
 - **Demurrage Reduction:** Decrease in vessel wait times and idle jetty occupancy.
 - **Operational Visibility:** 100% of vessel movements and QC/Quantity checks logged digitally.
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4. Scope

- **In-Scope:** Shipping Instruction visibility, Vessel allocation/berthing management, Loading/Unloading flow automation with SLA formulas, QC/Quantity check logging, Exception approval workflows, Jetty downtime management, and Master Data (Port/Jetty/SLA).
 - **Out-of-Scope:** Direct integration with vessel AIS (Global GPS) for real-time tracking (vessel position is manual update), Automated weather hardware (uses Google API instead).
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5. Business or User Flow

The system follows two primary paths: **Unloading (Purchasing)** and **Loading (Sales)**.

The SLA Formula

The system automatically calculates the Estimated Completion for activities using the following logic:

$$SLA = Q_1 + Q_2 + C + \left(\frac{\sum V_n}{Standard\ Rate \times Buffer} \right) + ((n - 1) \times S)$$

Equation	Description
Q_1	Quality & Quantity checking
Q_2	Final Quality & Quantity checking
C	Clearance
$\sum V_n$	The total sum of all material volumes.
<i>Standard Rate</i>	the "Planned Speed" at which material moves through your pipes or loading arms.
<i>Buffer</i>	configurable variable (default 0.85) that can be adjusted based on historical vessel performance
n	The number of different material types.
S	A fixed penalty (suggested 1 hour) for cleaning, flushing, or re-connecting hoses. <i>Why? when different materials share the same pipeline and require "pugging" or flushing in between.</i>

Note. All variables are configurable in the system.

6. User Stories

No	Function	User Story	Acceptance Criteria
1	Shipping Instruction	As a Jetty Operator, I want to view the list of incoming Shipping Instructions (SI) so I can see vessel, material, and purpose details.	• Display SI from Logistics & EXIM. • Filter by purpose (Loading/Unloading).
2	Allocation	As a Jetty Operator, I want to update vessel positions and ETB/ETA so that the berthing sequence is accurate.	• Log arrival update functionality. • Set berthing sequence and jetty assignment.
3	Docking (Timer)	As a Jetty Operator, I want to start a timer upon docking so the system calculates the Estimated Completion.	• Timer start triggers SLA formula calculation. • Variables (Standard Rate, Buffer, S) are configurable.
4	QC / Survey	As a QC Team member, I want to log survey results and upload documents so the Jetty team knows when to start work.	• Input field for survey status. • Mandatory document upload before next stage.

No	Function	User Story	Acceptance Criteria
5	Quantity Check	As a Tank Farm/Jetty Operator, I want to key in quantity results (sounding) so that loading progress is tracked.	• Entry for quantity check result. • Integrated into the Loading/Unloading flow.
6	Clearance & Sign-off	As a Jetty Operator, I want to sign off a vessel for sailing only when completion is 100%.	• Validation: 100% completion required for Sign-off. • Exception process requires justification & admin approval.
7	Dashboard	As a Manager, I want to see weekly trends (occupied jetties, demurrage, incidents) and weather.	• Integration with Google Weather API. • Charts for Weekly trends (Incoming/Sailing/Demurrage).
8	Jetty Ops	As a Jetty Operator, I want to set jetty status (usable vs. downtime) so planning is realistic.	• Status toggle: Available, Maintenance, High-Priority Booking.

7. Master Data & Platform

No	Feature	Requirements
7.1	Port & Jetty Master	Create master data for Sites and Jetty Specifications per Port.
7.2	SLA & Rate Master	Define Standard Rates and Target KPIs for Loading/Unloading per material type.
7.3	RBAC	Permissions at Department, Page, and Field level (View/Edit/Delete).
7.4	Audit Trail	Full logging of every field change, status update, and user login.

8. Dependencies / Risks

No	Dependency / Risk	Raised Date	Mitigation	PIC	Status
1	Data Stream Integration	09/03/2026	Ensure EXIM/Logistics data format matches Jetty SI requirements.	IT Lead	Open
2	Formula Accuracy	09/03/2026	Test SLA formula with historical data to calibrate "Buffer" defaults.	PPIC	Open
3	Connectivity	09/03/2026	Ensure the Jetty portal is accessible via mobile/tablets for outdoor operators.	IT Lead	New